The Human Heart

08/06/2026

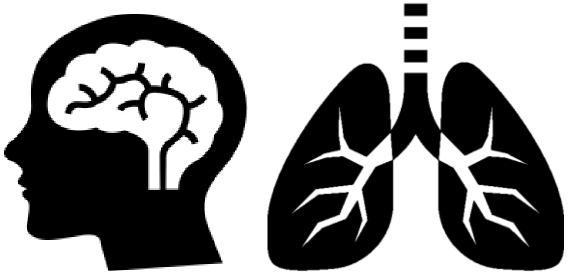
Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, aorta,

Starter: Choose the correct word to complete the sentence from each box. Write the sentences into your book.

EXT: Write down the key words for this lesson.

The heart is positioned at an angle on the **LEFT/RIGHT** side of your body

The heart sends blood to two places at the same time. The **BRAIN/LUNGS** and the body



The pulmonary vein is the only vein to carry **OXYGENATED/ DEOXYGENATED** blood

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Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, aorta,

Starter: Complete your pre-knowledge sheet about the heart.

EXT: Make a sentence using one or some of the key words.

	YES	NO
Heart is a muscle	☐	☐
The other name for heart tissue is 'cardiac'	☐	☐
The heart has 3 chambers	☐	☐
The heart pumps blood to two places	☐	☐
The heart does not have valves	☐	☐
The heart is an organ system	☐	☐
The bottom part of the heart is called 'atrium'	☐	☐
The top part of the heart is called 'ventricles'	☐	☐
The pace-maker is found in the 'right atrium'	☐	☐

	YES	NO
Heart is a muscle	✓	
The other name for heart tissue is ‘cardiac’	✓	
The heart has 3 chambers		✓
The heart pumps blood to two places	✓	
The heart does not have valves		✓
The heart is an organ system		✓
The bottom part of the heart is called ‘atrium’		✓
The top part of the heart is called ‘ventricles’		✓
The pace-maker is found in the ‘right atrium’	✓	

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Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, aorta,

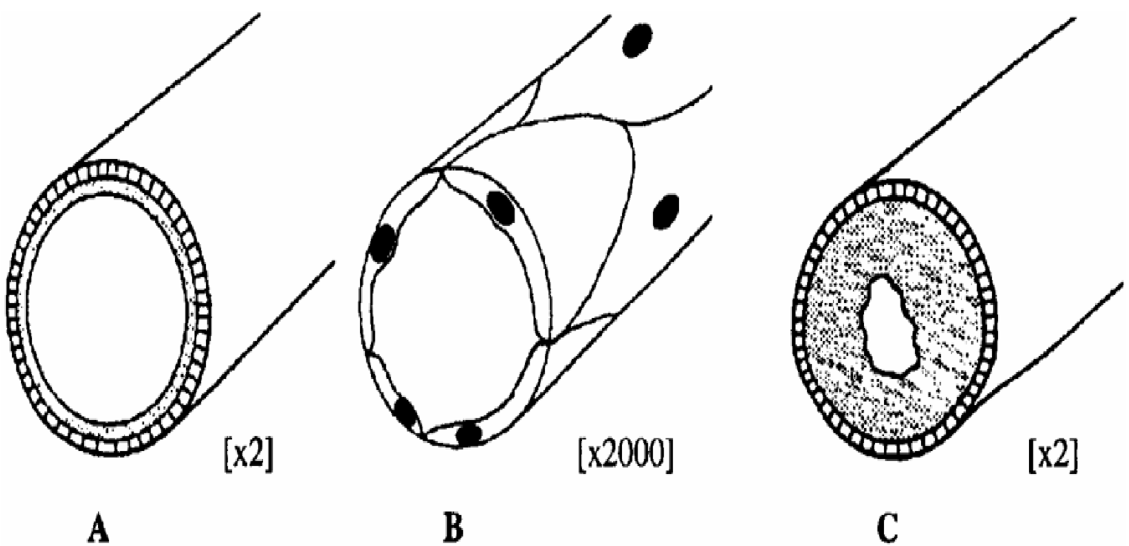
Starter: Answer these three review questions below.

EXT: What factors would affect how well the enzyme will work?

2. Name the three types of blood vessel and give a feature of each. [3]

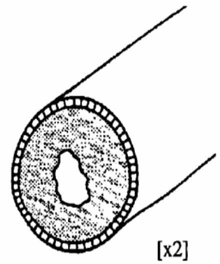
3. Explain the structure of the veins and how they relate to their function. [4]

1. Which vessel is picture C. What is its function? [2 marks]



Review

1. Which vessel is picture C. What is its function? [2 marks]



Artery. Send blood away from the heart at high pressure.

2. Name the three types of blood vessel and give a feature of each. [3 marks]

Artery, capillaries, veins
Arteries – thick muscle, elastic walls
Capillaries – thin and narrow
Veins – valves, wide lumen

3. Explain the structure of the veins and how they relate to their function.

Veins are elastic, wide lumen vessels with valves. The valves stop blood flowing backwards so low pressure blood can return back to the heart. The lumen is large to carry more blood.

The Human Heart

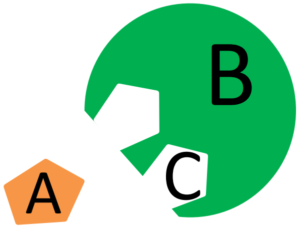
08/06/2026

Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, aorta,

Starter: Complete these quiz questions about last lesson

Quiz 6: How the Digestive system works

1. Pepsin breaks down a) Carbonhydrate b) Lipids c) Protein
2. Amylase breaks down a) Carbonhydrate b) Lipids c) Protein



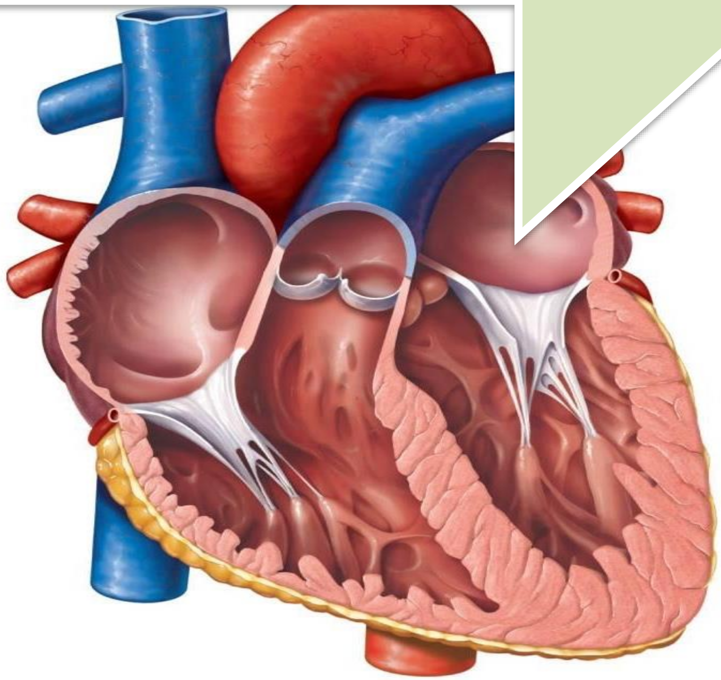
3. Amylase is found in a) Stomach b) Saliva c) Small intestine
4. Carbohydrates break down into a) Glucose b) Amino acids
5. Lipase is found in a) Stomach b) Mouth c) Small intestine
6. What part on the diagram is the substrate? A, B or C
7. What part is the active site A, B or C
8. What do root hair cells absorb using active transport
a) water b) Mineral ions c) glucose d) soil
9. Active transport moves substances from a ______ concentration to a _______ concentration across a semi permeable ___________
10. Which is found in bacterial cells but not found in plant cells
a) cytoplasm b) nucleus c) plasmids d) mitochondria

LOb: Understand the components of the heart.

Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, Coronary heart disease

Learning Outcomes:

- Describe the structure and function of the heart
- Explain how coronary heart disease develops and its risk factors
- Evaluate methods to prevent CHD



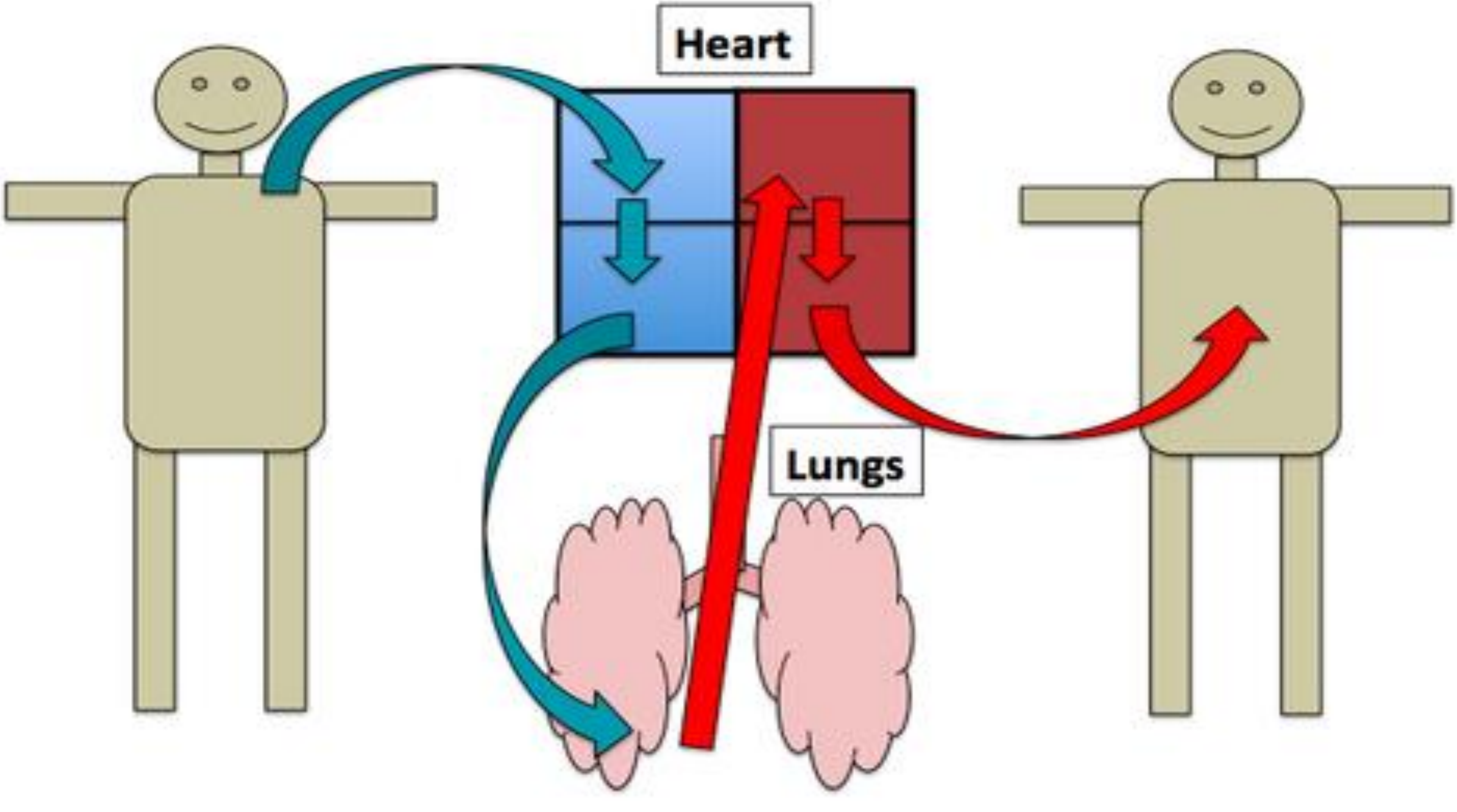
Did you know… your heart will beat around 4,800 times an hour every hour for the rest of your life. Your heart needs to be strong to keep that going for that long.

Heart Facts #1

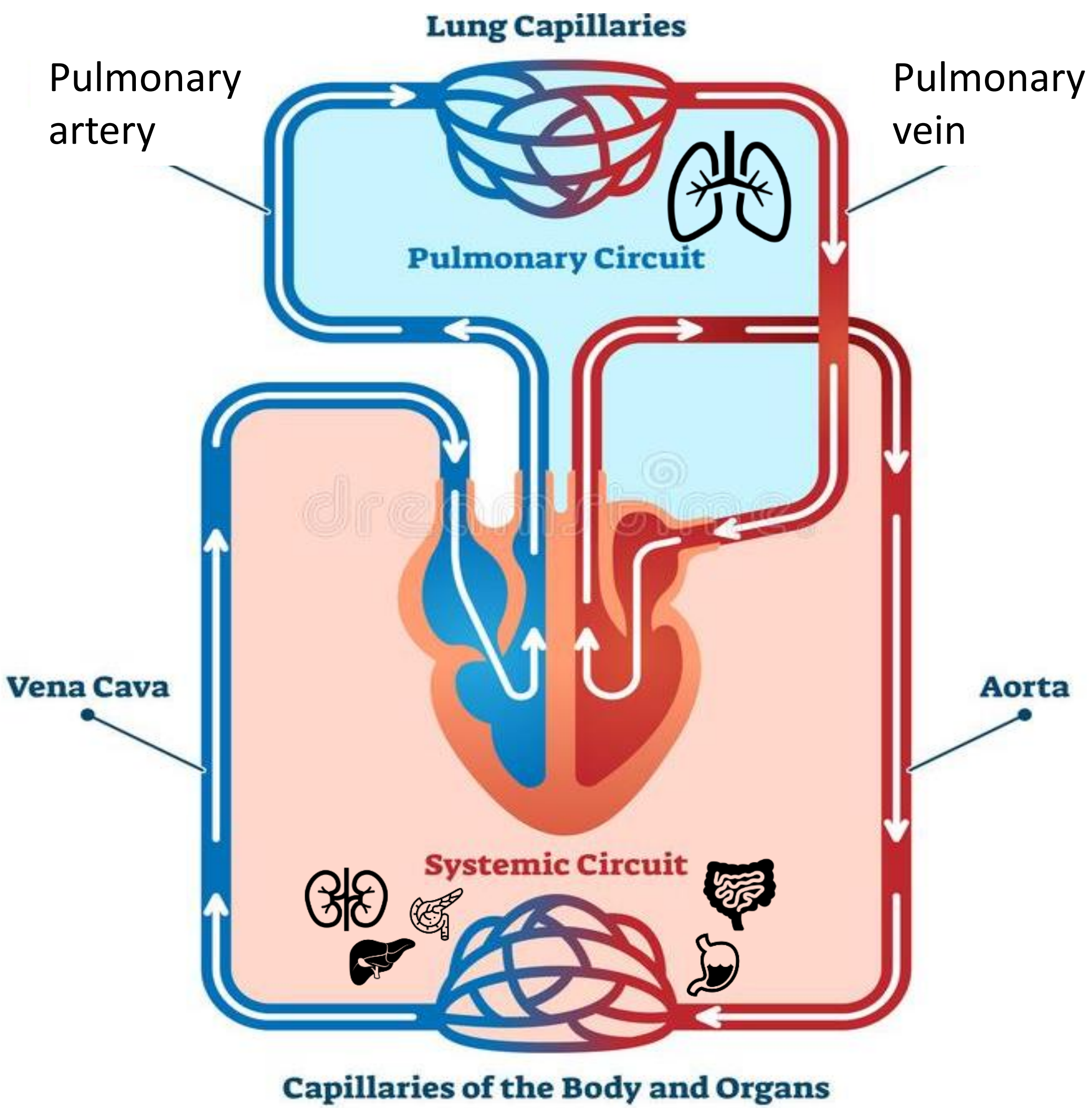
The heart is a muscle. It has a special name, **'Cardiac muscle'.**

Heart Facts #2 Double Pump

There are two sides to the heart. Each side pumps blood to a separate place.



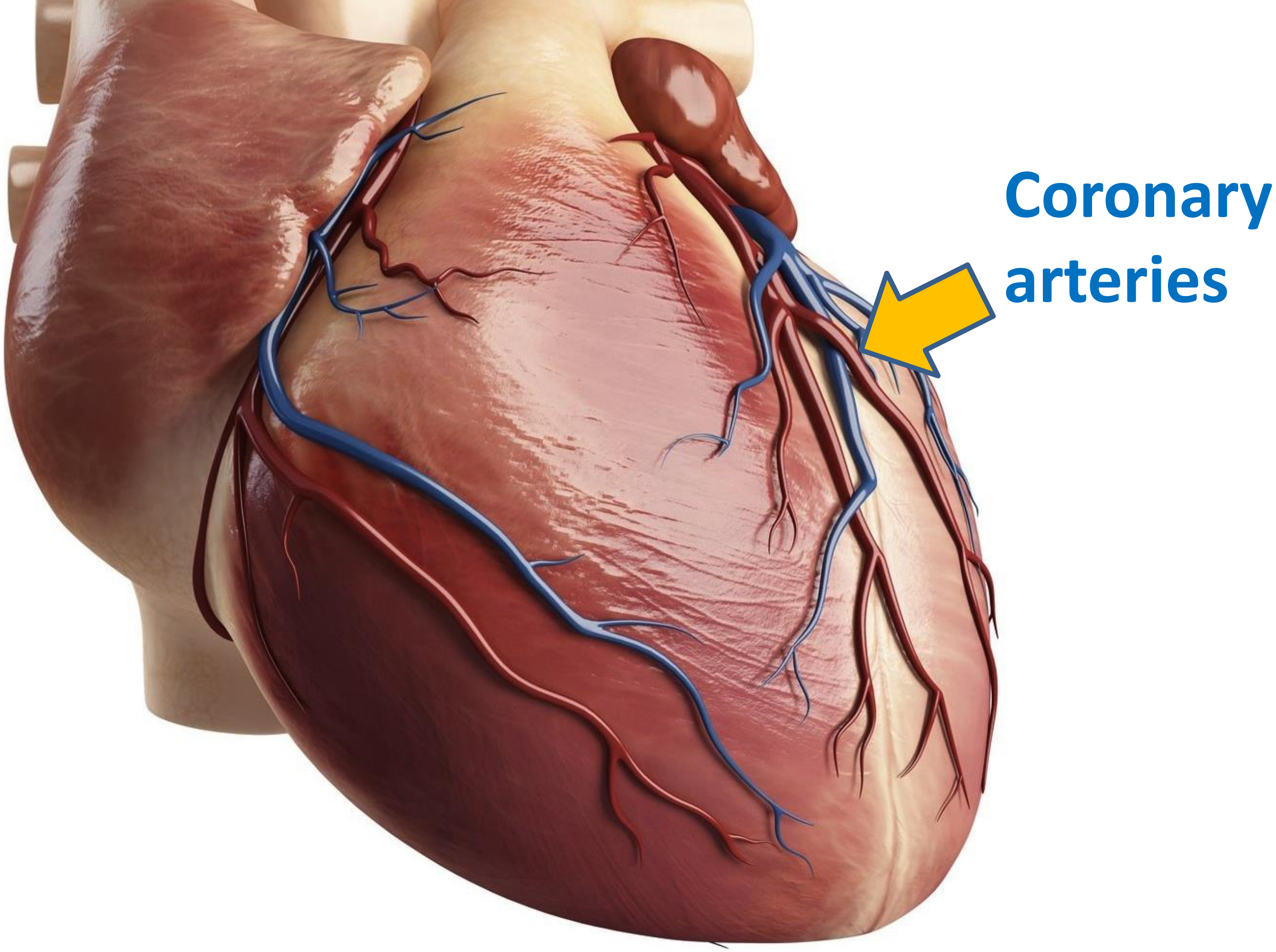
Double circulatory system. Why is it called Double?



Allows **blood** to be **sent** at **very high pressure**

Makes the circulatory system **very efficient**

More areas of the body receive oxygenated blood

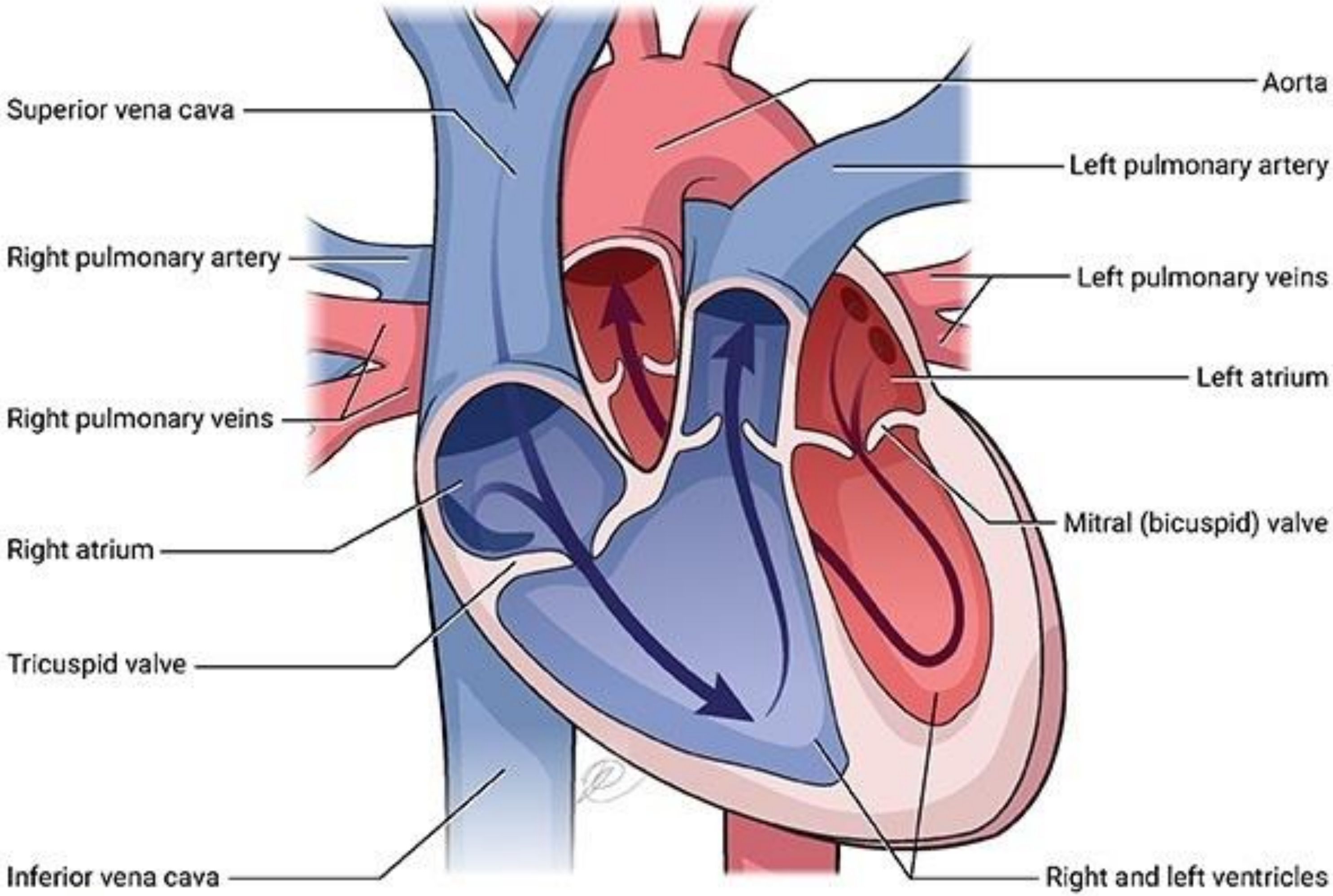


Heart Facts #3 Excellent blood supply

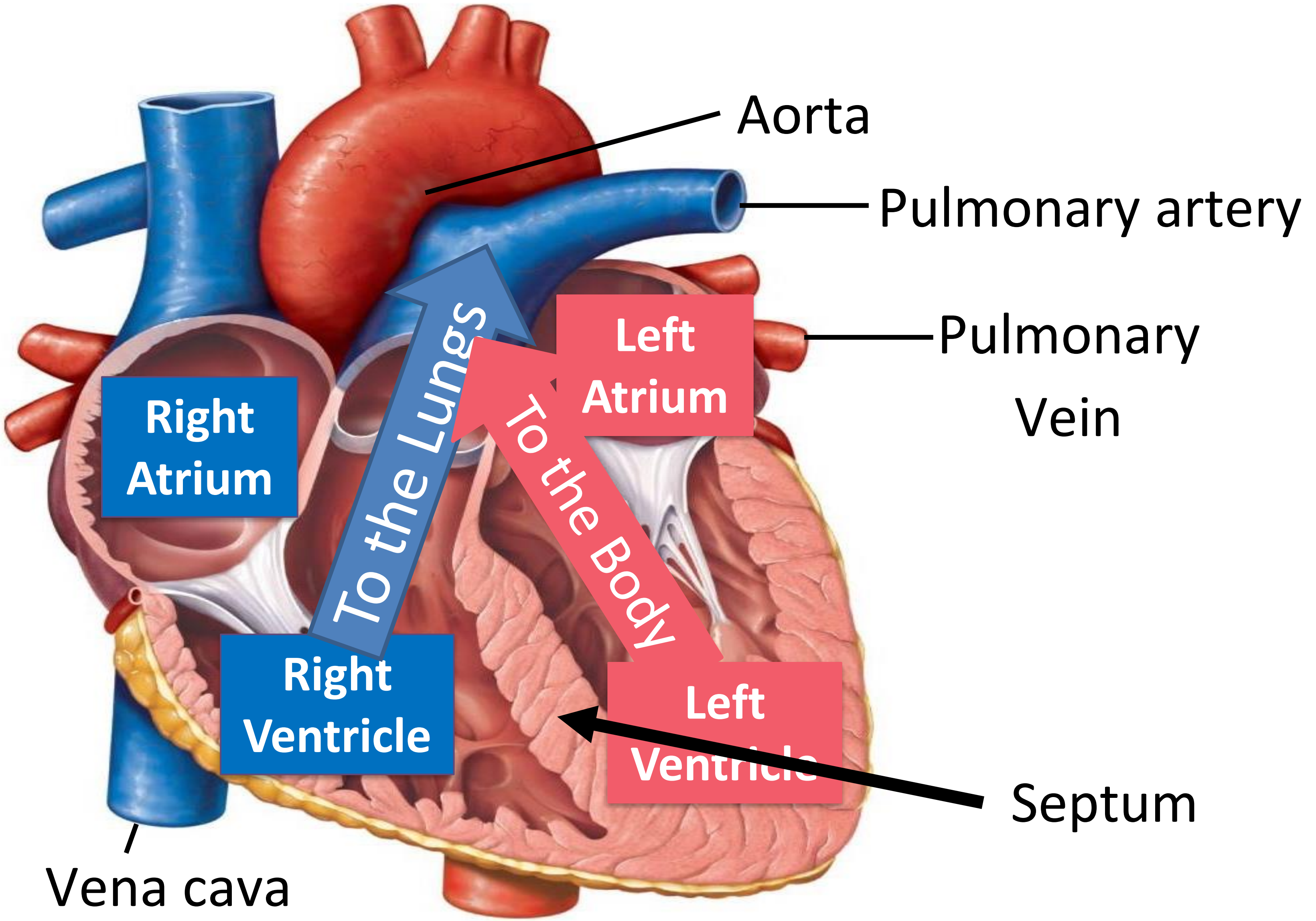
The **coronary arteries** supply the heart muscle with blood that it needs.

Heart Fact #4

Misconception: There is no blue or red side. They are what artists uses to show which side of the heart carries oxygenated blood (red) and deoxygenated blood (blue)



The structure of the heart



Activity: Label your heart with the following structures

KEYWORDS

Pulmonary artery

Left atrium

Right atrium

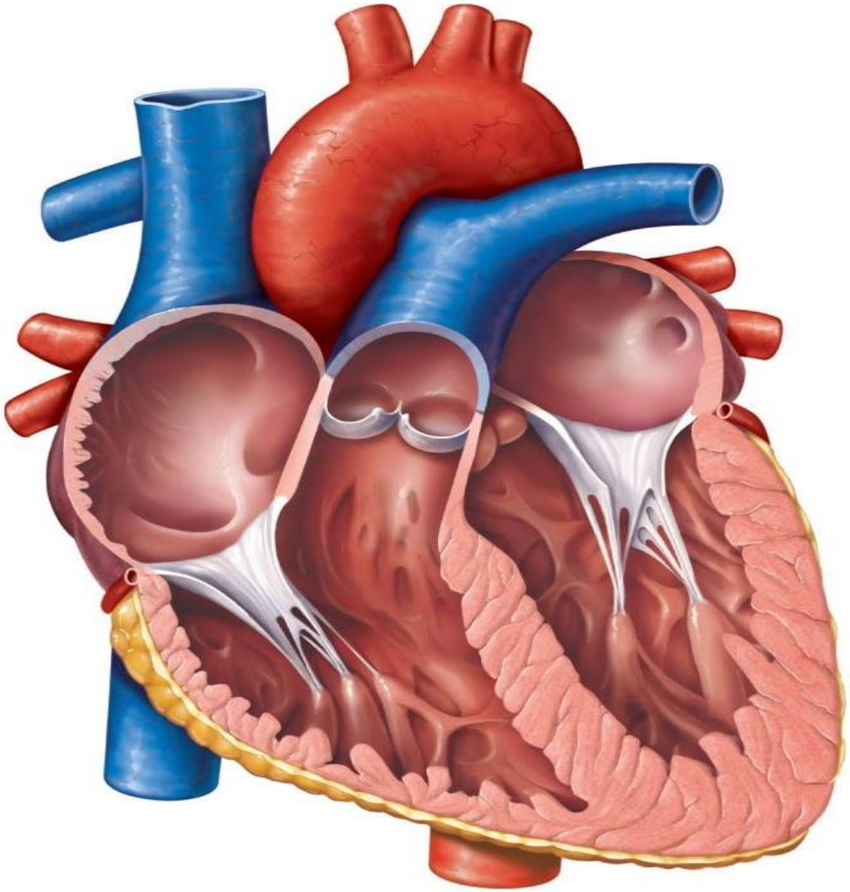
Aorta

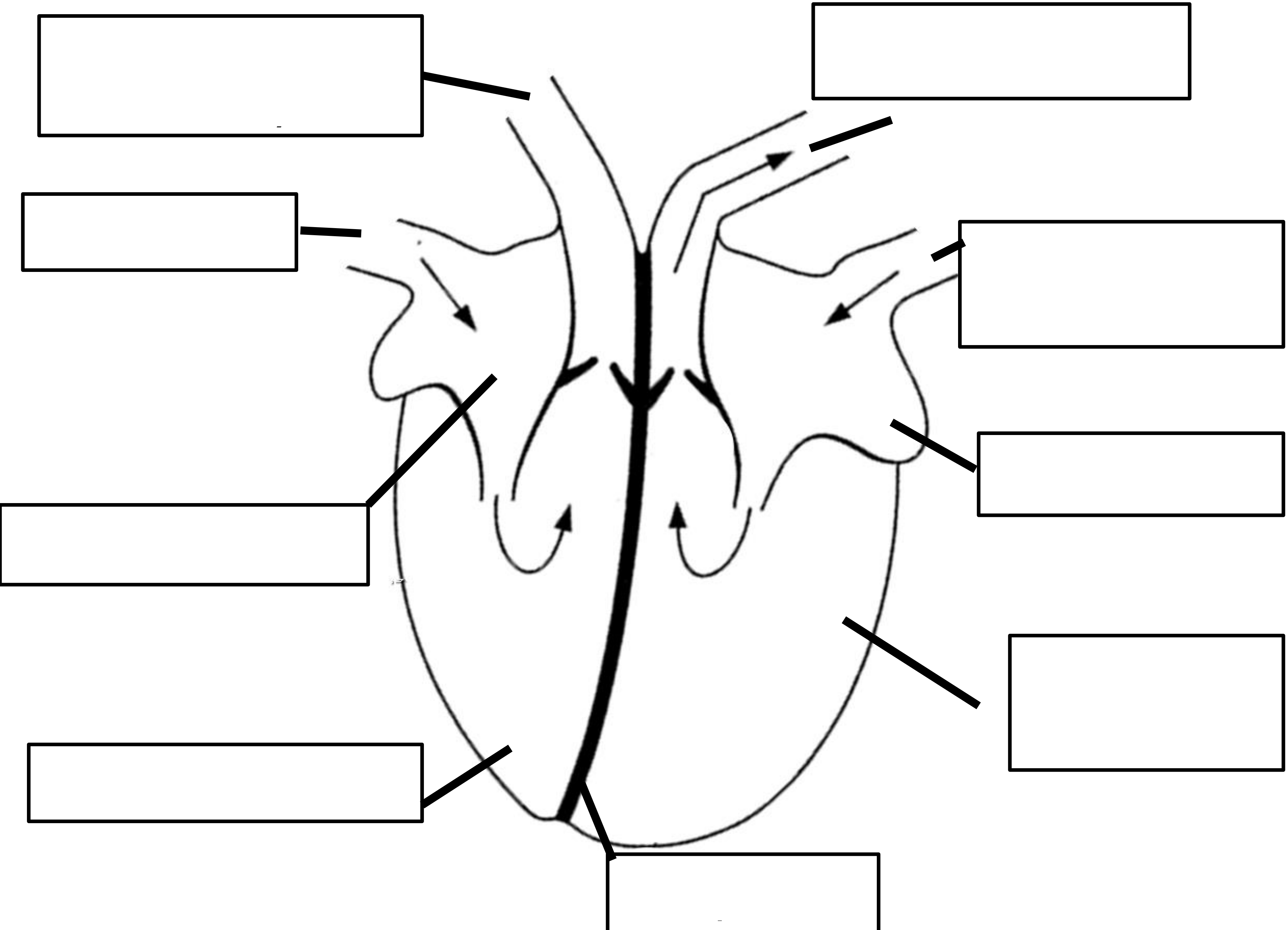
Pulmonary vein

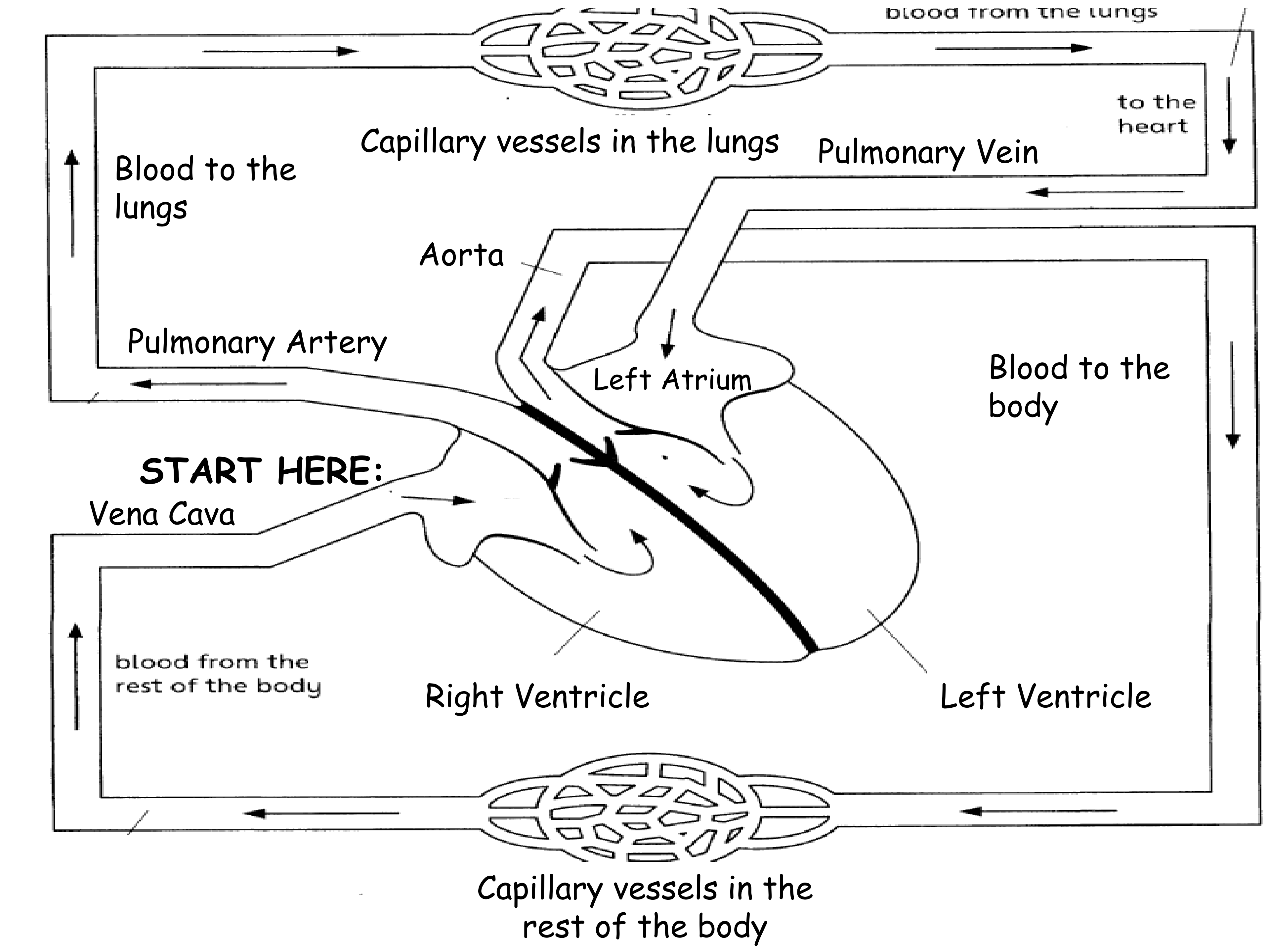
Septum

Left ventricle

Right Ventricle







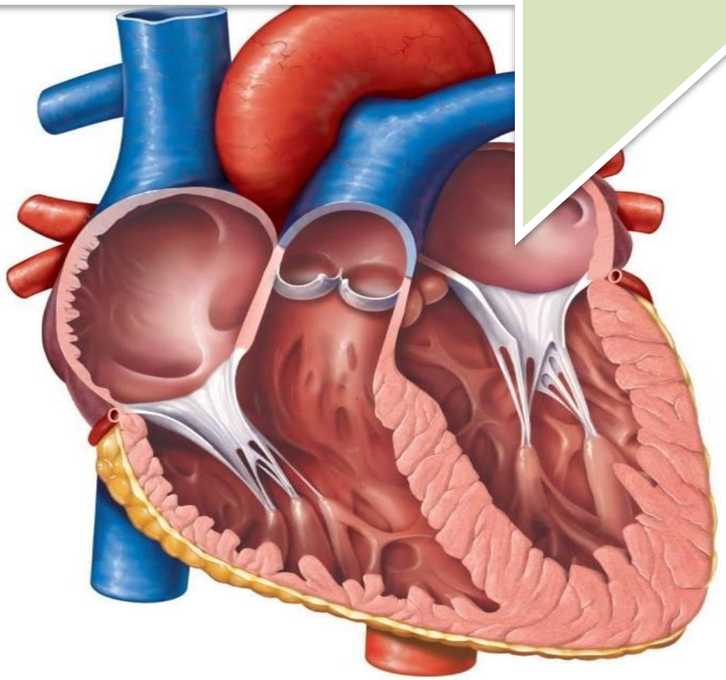
blood from the lungs
Capillary vessels in the lungs
to the heart
Pulmonary Vein
Blood to the lungs
Aorta
Pulmonary Artery
Left Atrium
Blood to the body
START HERE:
Vena Cava
blood from the rest of the body
Right Ventricle
Left Ventricle
Capillary vessels in the rest of the body

Activity: Describe the journey of blood as it enters the heart on its way to the lungs.

Blood enters the _ Right _ atrium. It then travels to the Right ___ Ventricle ____. It makes its journey to the lungs through the pulmonary __ Artery ____. After collecting oxygen from the lungs it makes its way back to the heart through the pulmonary ___ Vein ___. It enters the left _ Atrium __ first and then the left ___ Ventricle ____ before going to the body through the ___ aorta ___. Oxygen is used up by the body and blood returns to the heart via the _ Vena _ _ cava __

LOb: Understand the components of the heart.

Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, Coronary heart disease

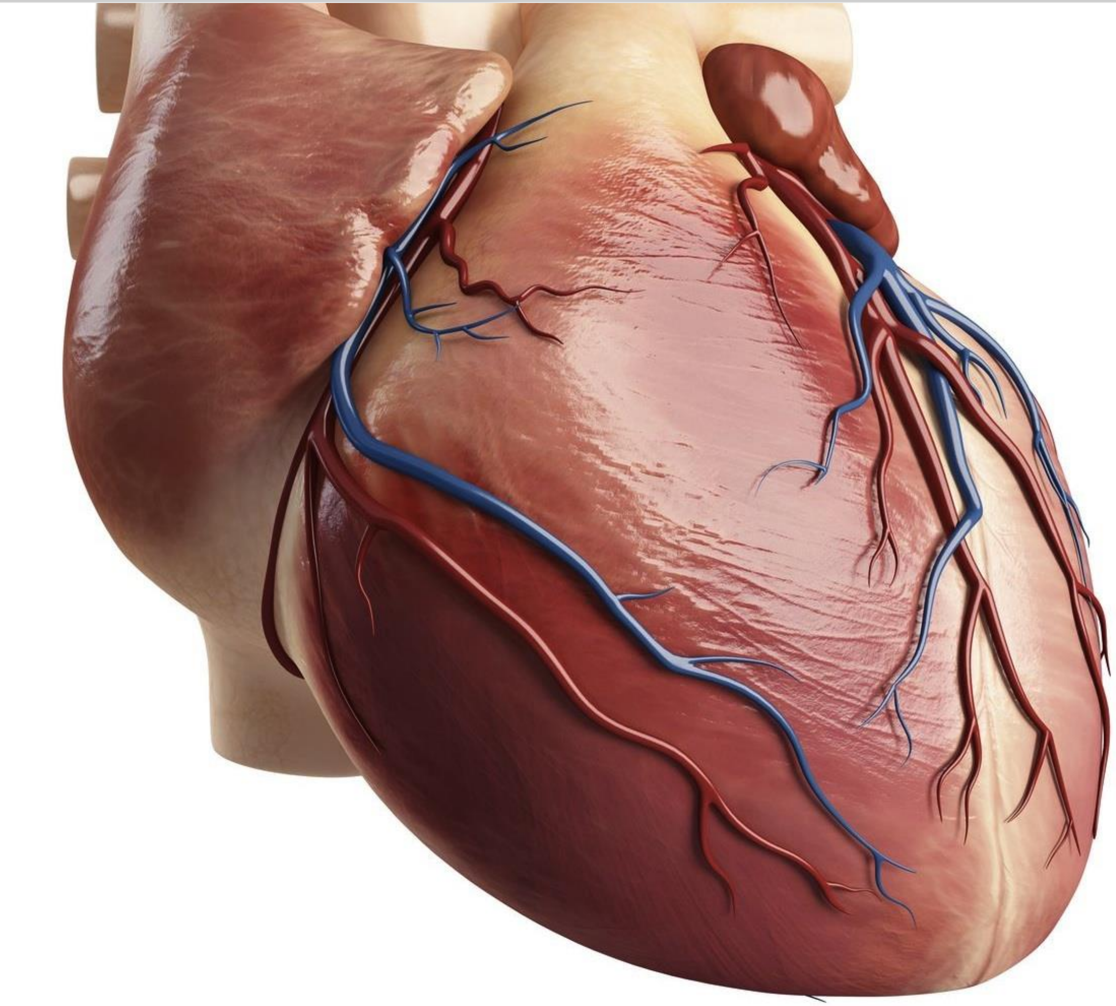


Learning Outcomes:

- ✓ Describe the structure and function of the heart
- ➢ Explain how coronary heart disease develops and its risk factors
- ➢ Evaluate methods to prevent CHD

Did you know… your heart will beat around 4,800 times an hour every hour for the rest of your life. Your heart needs to be strong to keep that going for that long.

Cardiovascular disease is a name to describe diseases that affect the blood vessels of the heart. **Coronary heart disease (CHD)** is a **very common** disease affecting the coronary arteries.



There are two types of diseases

Communicable

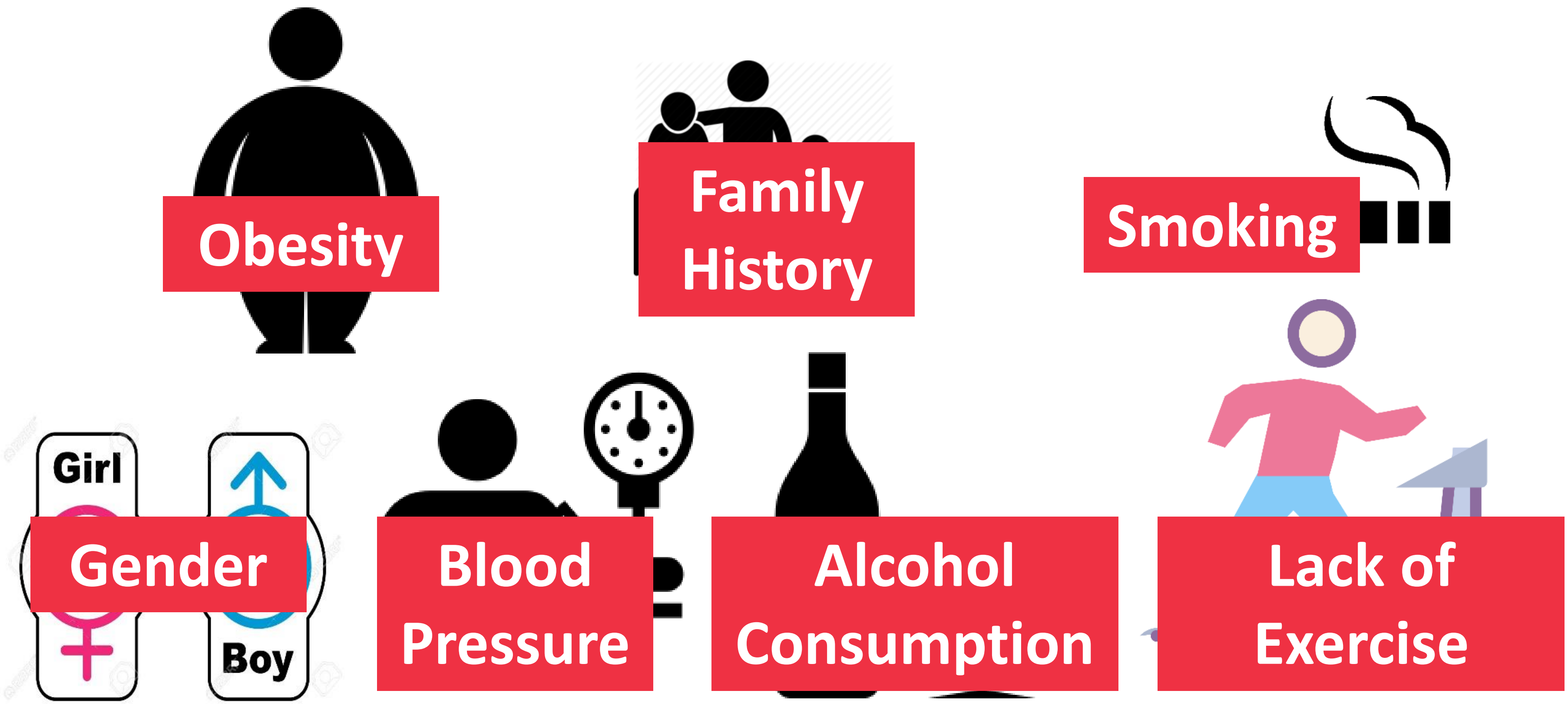
Non- Communicable

Communicable diseases are spread/passed on from one organism to another by pathogens. Cardiovascular disease is non-communicable.

What are the causes of Communicable disease?

Communicable diseases are caused by four main things: *VIRUS, BACTERIA, FUNGI* and *PROTISTS*

There are factors that make it more likely to have this disease these are called risk factors. Make a list using the pictures below.

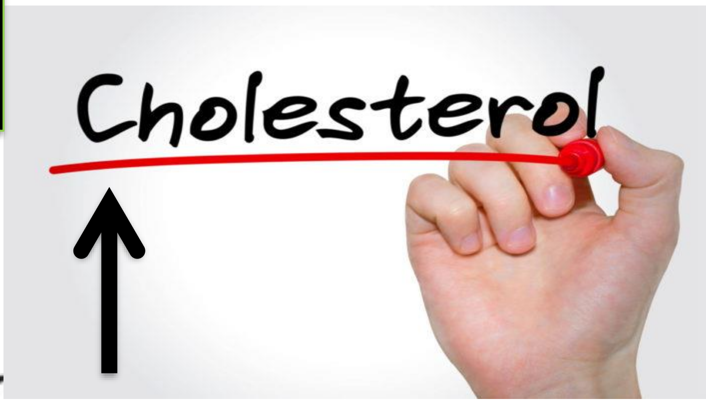


Watch this video about how coronary heart disease occurs and make notes about it.

Eating fatty foods and doing a low amount of exercise leads you down the road to having high cholesterol in your blood.

Cholesterol is a fatty substance needed for cells to make cell membranes and for growth.

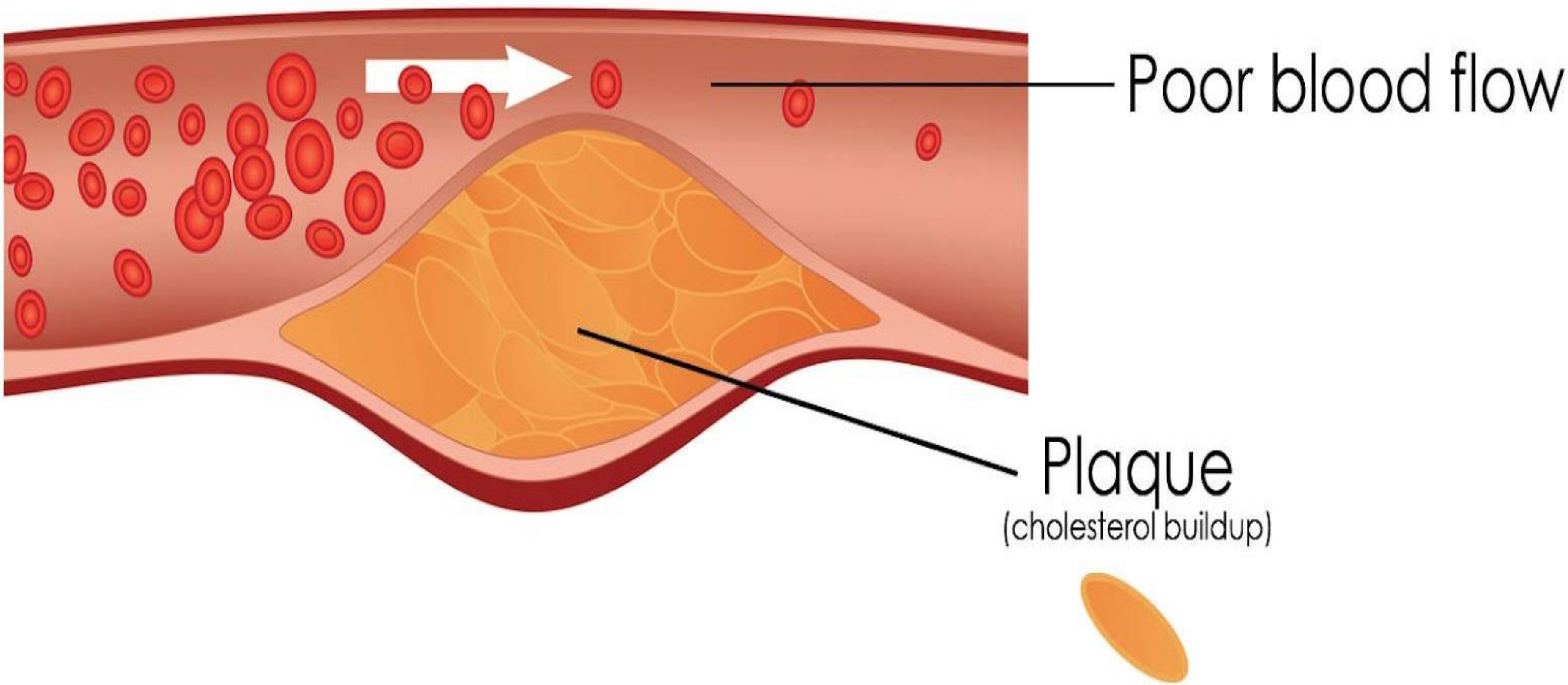
INCREASE



Fatty Foods

Lack of exercise

Over time the cholesterol will stick to the walls of your arteries and turn into a hard yellow substance called 'plaque'. This plaque can block your arteries stopping blood getting to your heart.

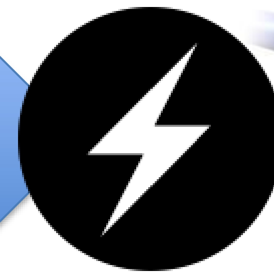


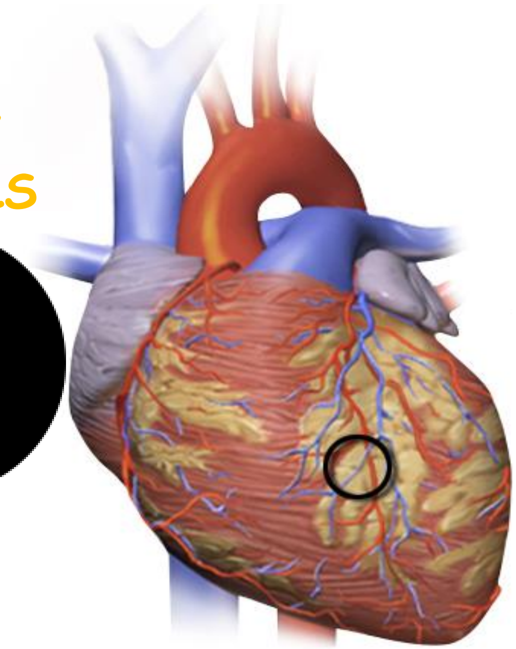
Why is it bad if the heart doesn't get enough blood?

RBCs contains oxygen
blood contains nutrients
(glucose)

Oxygen and glucose used for respiration.

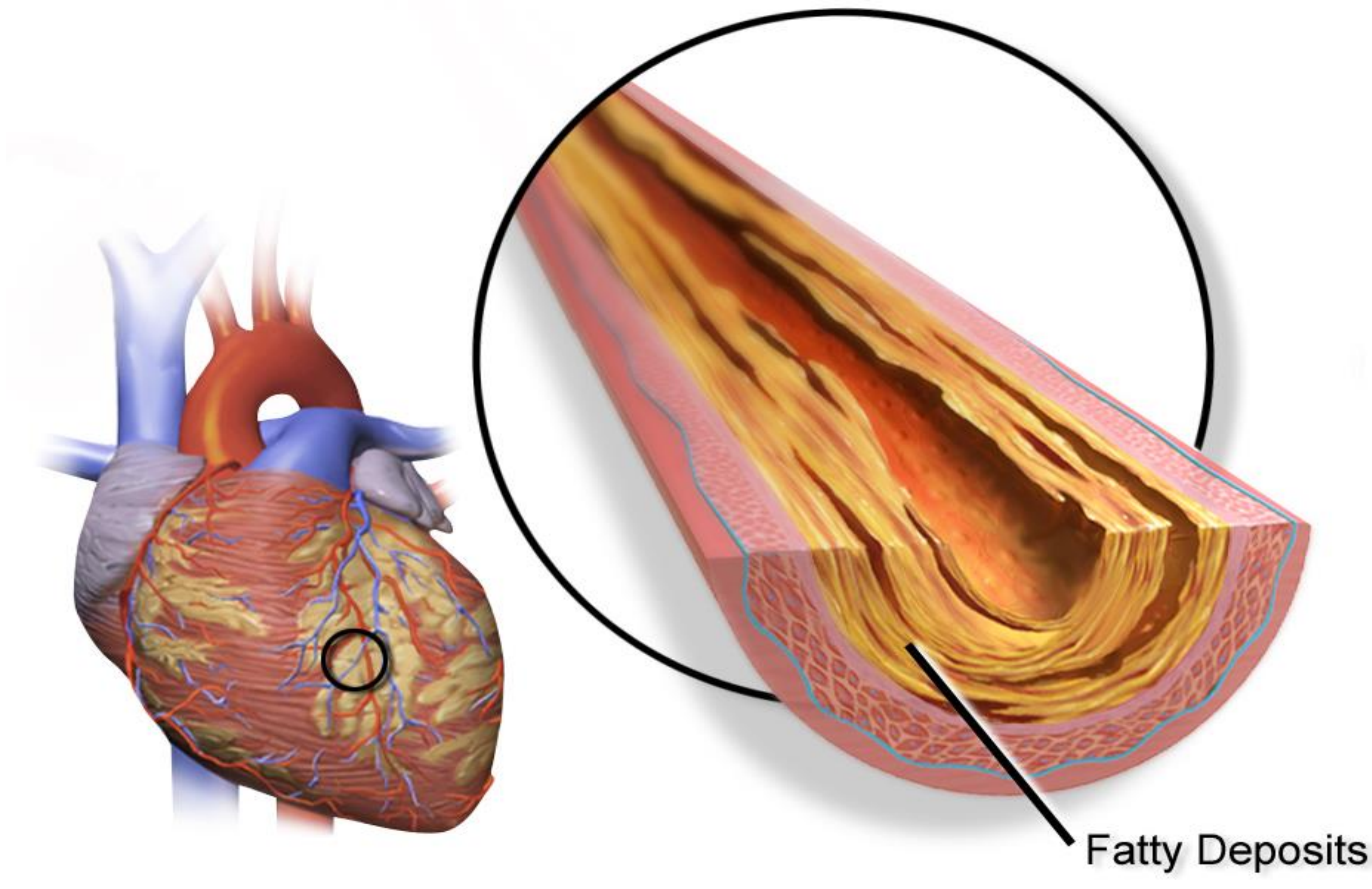
ENERGY
FOR CELLS





If the heart doesn't get blood, it doesn't get oxygen or glucose. Heart muscle cells can't release energy via respiration causing cells to die.

If this happens in the coronary arteries of the heart it could lead to a **heart attack (cardiac arrest).**

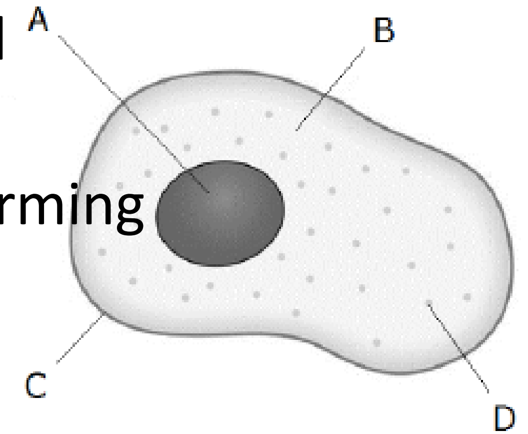


Quiz: CHD

1. Give one way to increase cholesterol in your blood

2. What arteries supply the heart muscle with blood
a) Conary b) Coro c) coronary d) cornary

3. Cholesterol builds up on the walls of the artery forming
a a) clot b) block c) blood flow d) plaque



4. Muscles need what two things to respire?
a) Glucose b) carbon dioxide c) water d) oxygen

5. What substance does red blood cells carry?

6. Enzymes are biological a) fats b) catalysts c) substrates d) amino acids

7. What type of cell has plasmids a) eukaryotic b) prokaryotic

8. Where are adult stem cells found a) blood b) brain c) bone marrow

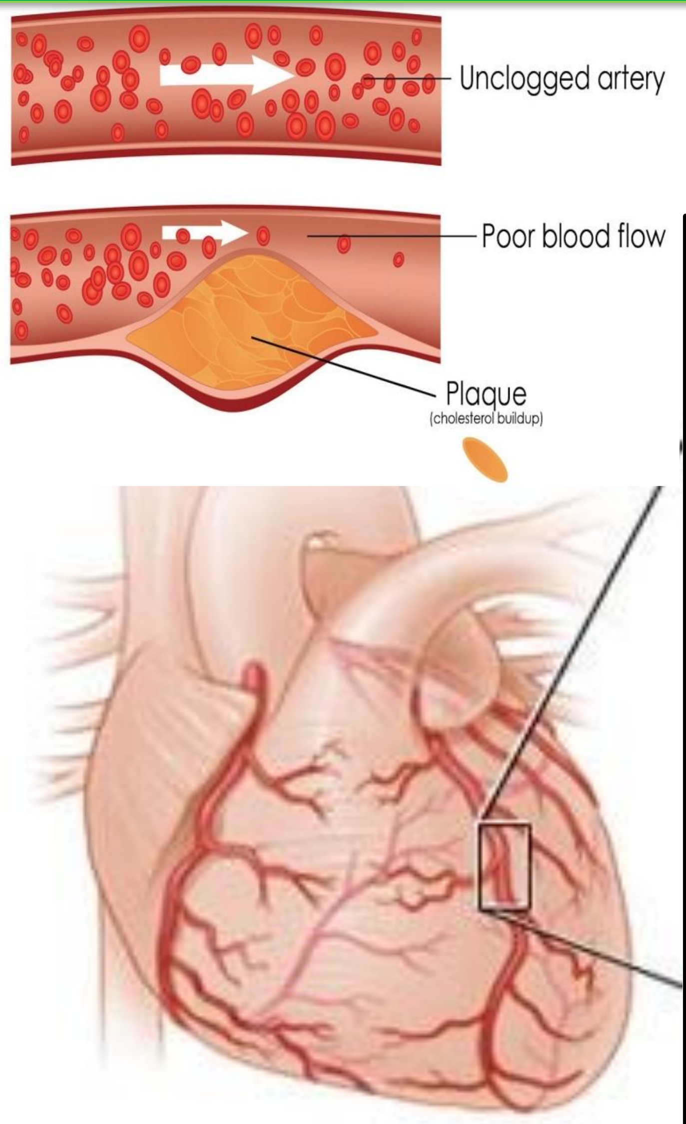
9. Some potato was placed into salty water what would happen
a) Mass is lost by osmosis b) mass is lost by diffusion c) mass gained by osmosis d) mass gain by diffusion

10. Active transports happens in a) sperm cells b) root hair cells c) leaf cell

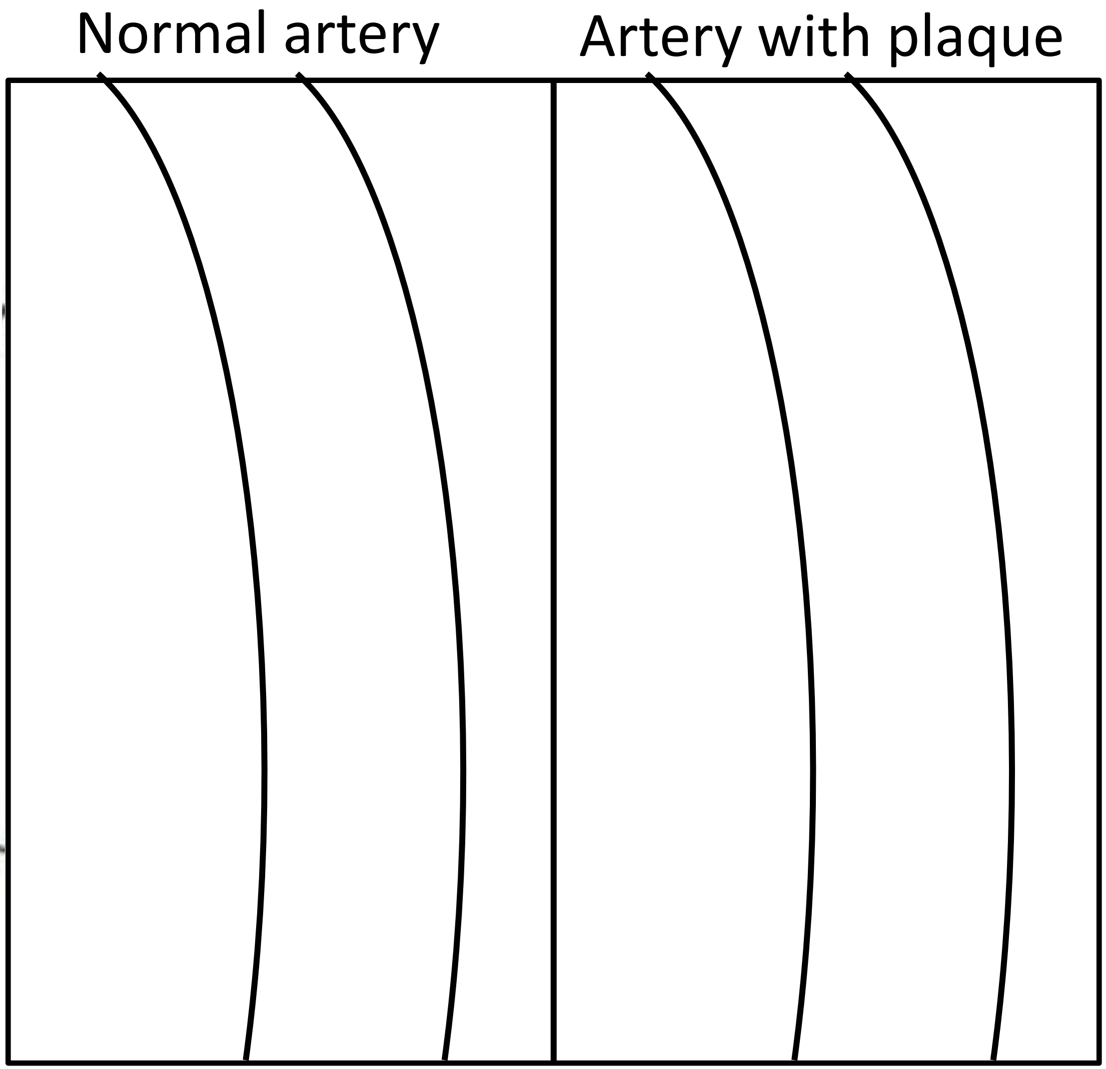
keyfacts

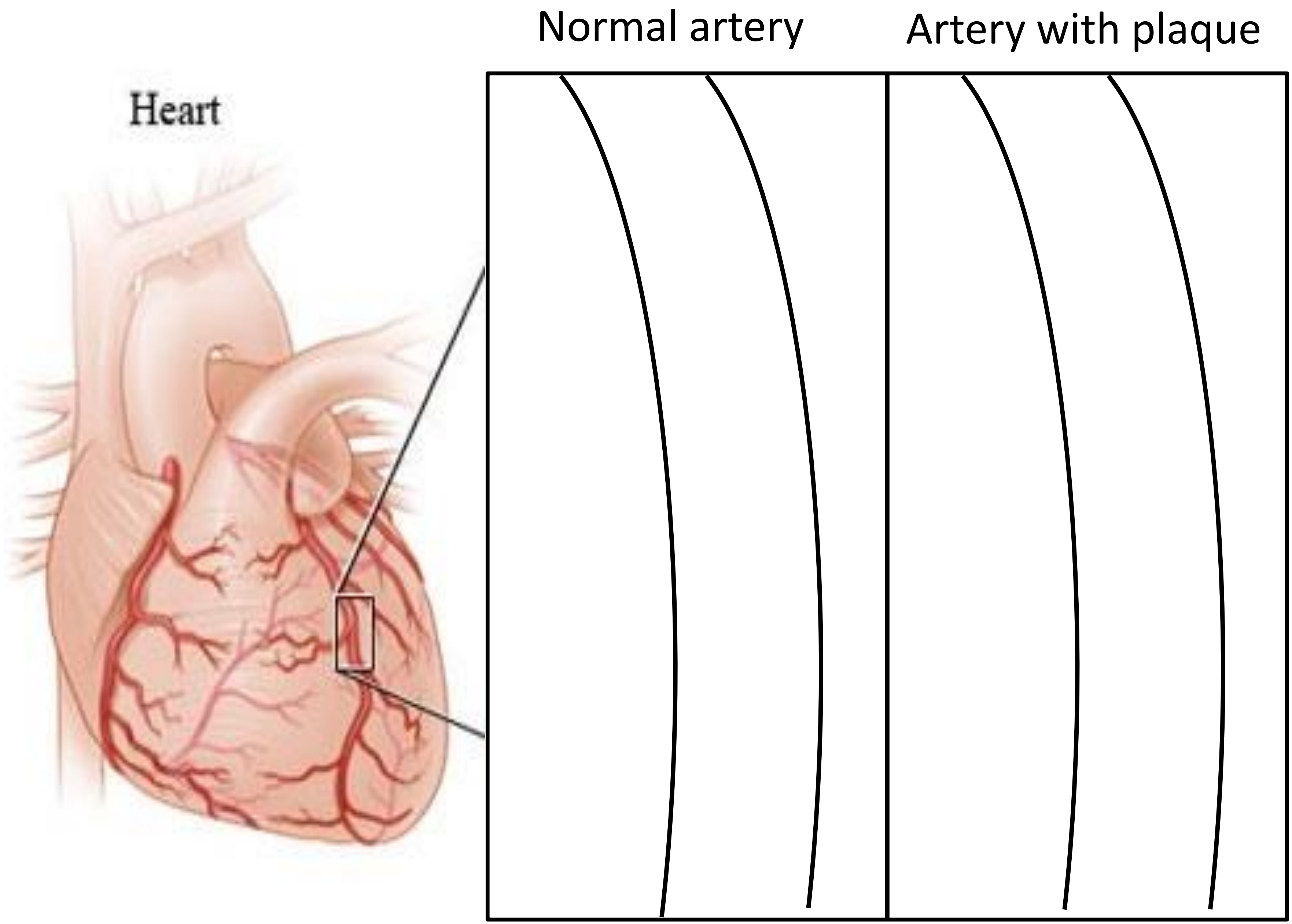
- Your heart has arteries called ‘coronary arteries’.
- Fatty food + Lacking exercise + Time = high cholesterol.
- Cholesterol builds up on your arteries and turns hard. this is called plaque.
- Plaque stops blood flow through your arteries. Less blood means less oxygen and glucose for cell respiration.
- Cell will begin to die and if enough cells die then a heart attack can happen.

Activity: Label your diagram and describe in your own words what coronary heart disease is.



Keywords: Plaque | cholesterol | arteries




Normal artery
Artery with plaque
Heart

Activity: Explain how CHD can cause a heart attack. Answer in your own words or using the structure below. [4 marks]

Coronary heart disease is when __________ builds up on the walls of the _________ artery. This then hardens and turns into a _______. The plaque _________ blood flow to the heart. This means less ________ reaches the heart muscle so they can not do __________ ___________. This means the cells do not release _________ so they die which causes a ________ _________

Review

Coronary heart disease is when cholesterol builds up on the walls of the coronary artery. This then hardens and turns into a plaque. The plaque reduces blood flow to the heart. This means less oxygen reaches the heart muscle so they can not do aerobic respiration. This means the cells do not release energy so they die which causes a heart attack

Review

Reduce/ restricted/ stopped blood flow [1]

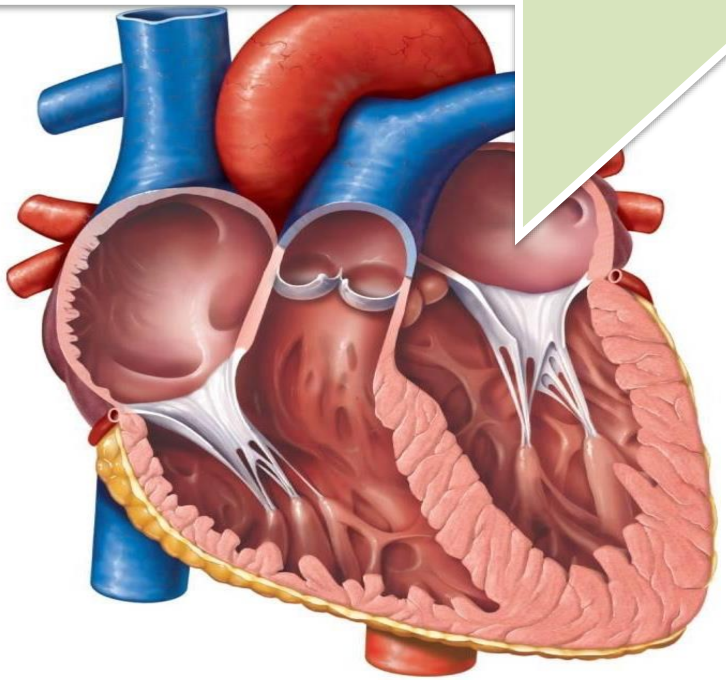
Less oxygen reaches the heart muscle cells [1]

So the heart muscle cells cannot respire [1]

So the cells do not releases energy [1]

LOb: Understand the components of the heart

Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, Coronary heart disease



Learning Outcomes:

- ✓ Describe the structure and function of the heart
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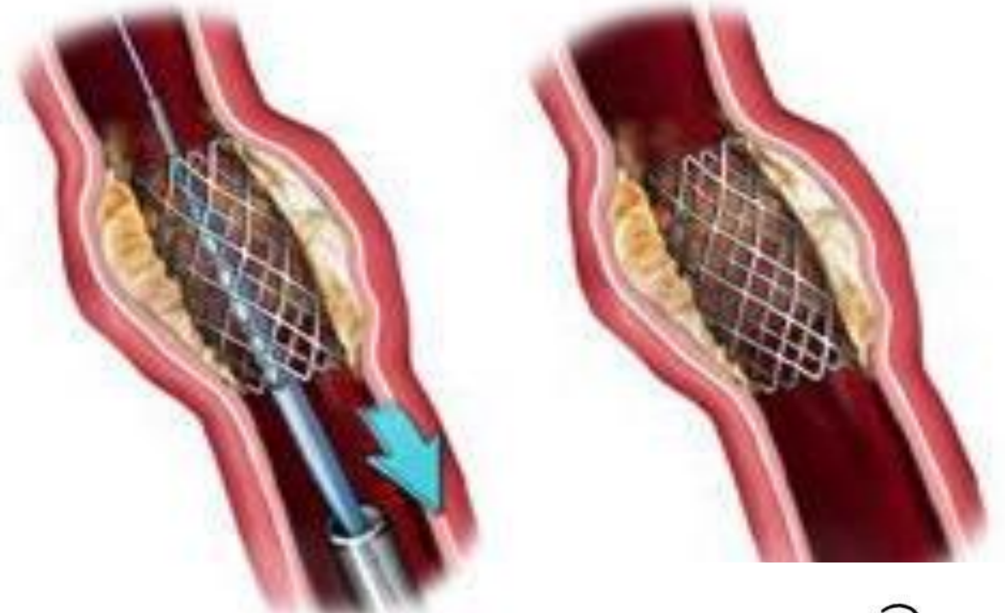
Did you know… your heart will beat around 4,800 times an hour every hour for the rest of your life. Your heart needs to be strong to keep that going for that long.

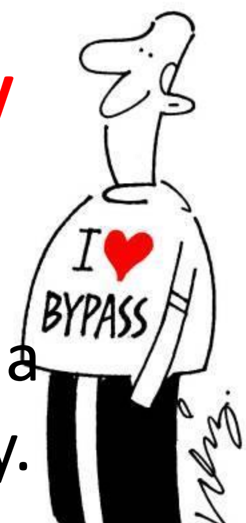
Preventing Coronary Heart Disease

1. A **stent** is a wire mesh that can expand to widen the artery.



3. A **statin** is a drug that lowers the cholesterol in our blood

2. **Heart by-pass surgery** is when you replace blocked or narrow arteries with veins from a another part of the body.



Method	Advantages	Disadvantages
Statins	Reduce blood cholesterol levels Proven to reduce the risk of CHD Non-invasive Inexpensive	Unwanted side effects (e.g.liver problems Have to remember to take pills
Stent	Good blood flow through the artery No general anaesthetics needed.	Doesn’t treat underlying cause Could lead to blood clot Risk of bleeding or further damage to the heart or vessel
Bypass Surgery	Treats badly blocked arteries	Risky of surgery complication General anaesthetic needed Expensive Does treat underlying cause

Use the information in the table to evaluate the best method for each person below. Explain advantages and disadvantages of the methods chosen.

45 year old Gary has a history of CDH in his family he worries about getting CHD himself.

50 year old Alberto had scans at hospital. He has a severally narrowed artery. He has had a surgery before. He is deemed too risky for another.

52 year old Clara has some narrow arteries and some severely narrowed arteries.

Review

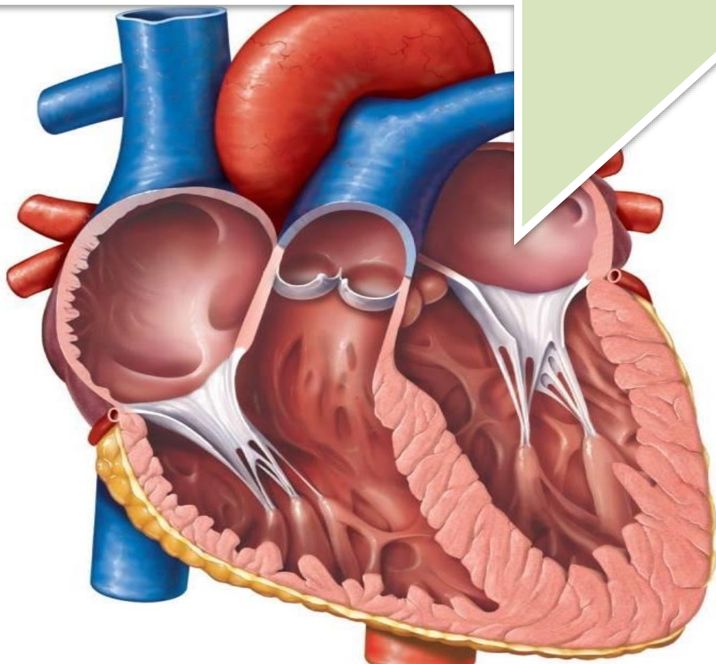
Give him statins. Monitor his health. Make sure he has good exercise levels and is improving his diet ADV: They are proven DIS: Could cause side effects

Also give him statins. No bypass surgery so give him a stent. ADV: No general anaesthetic DIS: Arteries could be too bad for a stent. Stent could damage heart and stent won’t treat underlying condition.

Also give her statins. Stent surgery and bypass surgery. ADV: Will allow better blood flow DIS: risk of surgery e.g. infection. Expensive.

LOb: Understand the components of the heart.

Keywords: Atrium, ventricles, valves, pump, chambers, cardiac muscle, coronary arteries, veins, Coronary heart disease



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keyfacts

- **Heart muscle** is called **cardiac muscle.** The heart pumps blood in a **double circulatory system.**

- The main **blood vessels of the heart** include the aorta, vena cava, pulmonary artery, pulmonary vein and coronary arteries.

- **Coronary heart disease** is caused by **plaque** build up. It **Reduces blood flow** which **decreases oxygen** and **glucose** getting to cells so they **can not do respiration leading to a HA.**

- **Stents, statin drugs** and **heart by-pass** used for the disease.

Plenary: Rate how you found the lesson. Lemon & herb for it was easy, Extra Hot for it was really hard. Then try a question.

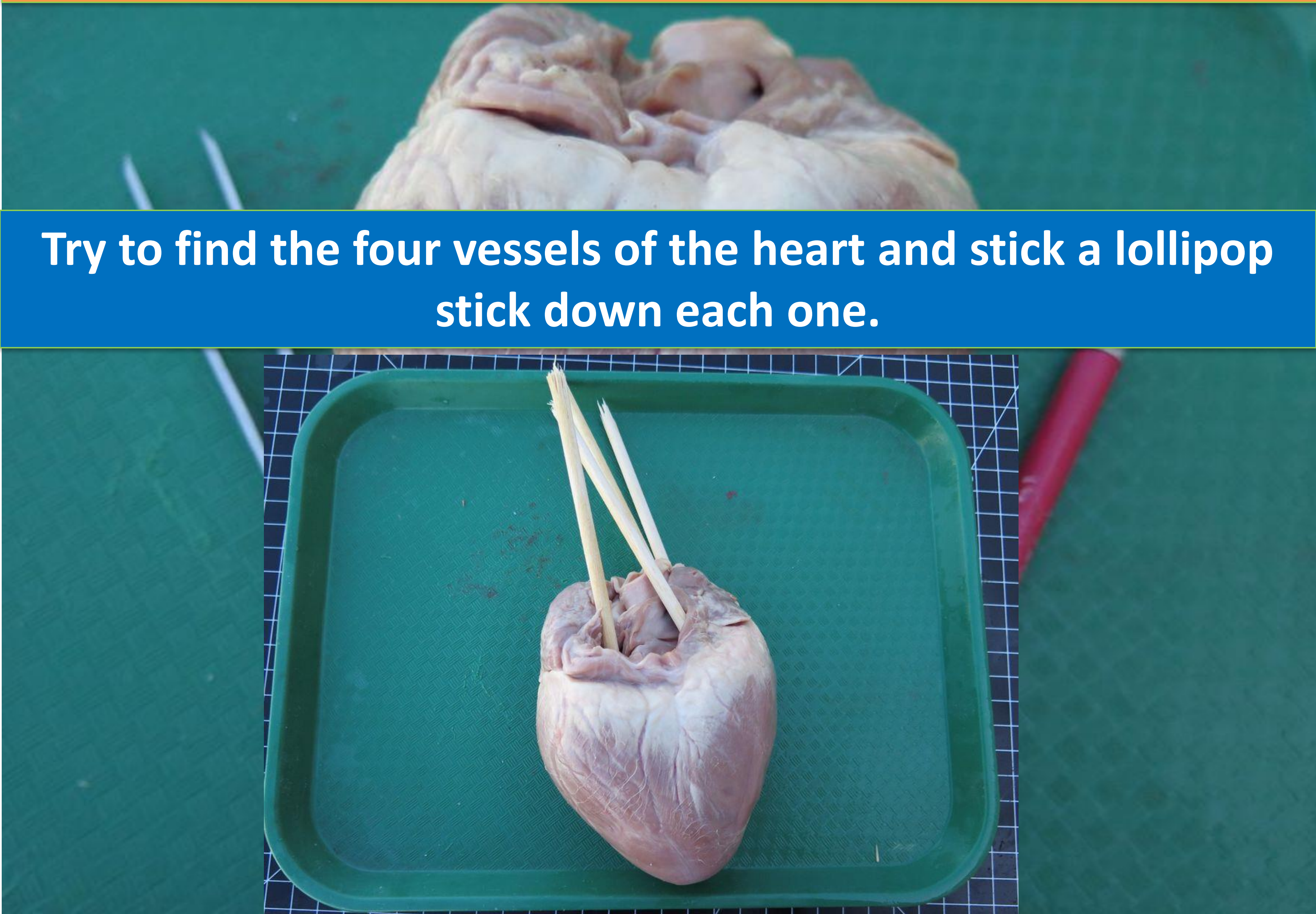
Describe which method of treatment for Coronary heart disease is the easiest but why is might be annoying over a long time.

Describe what a plaque is.

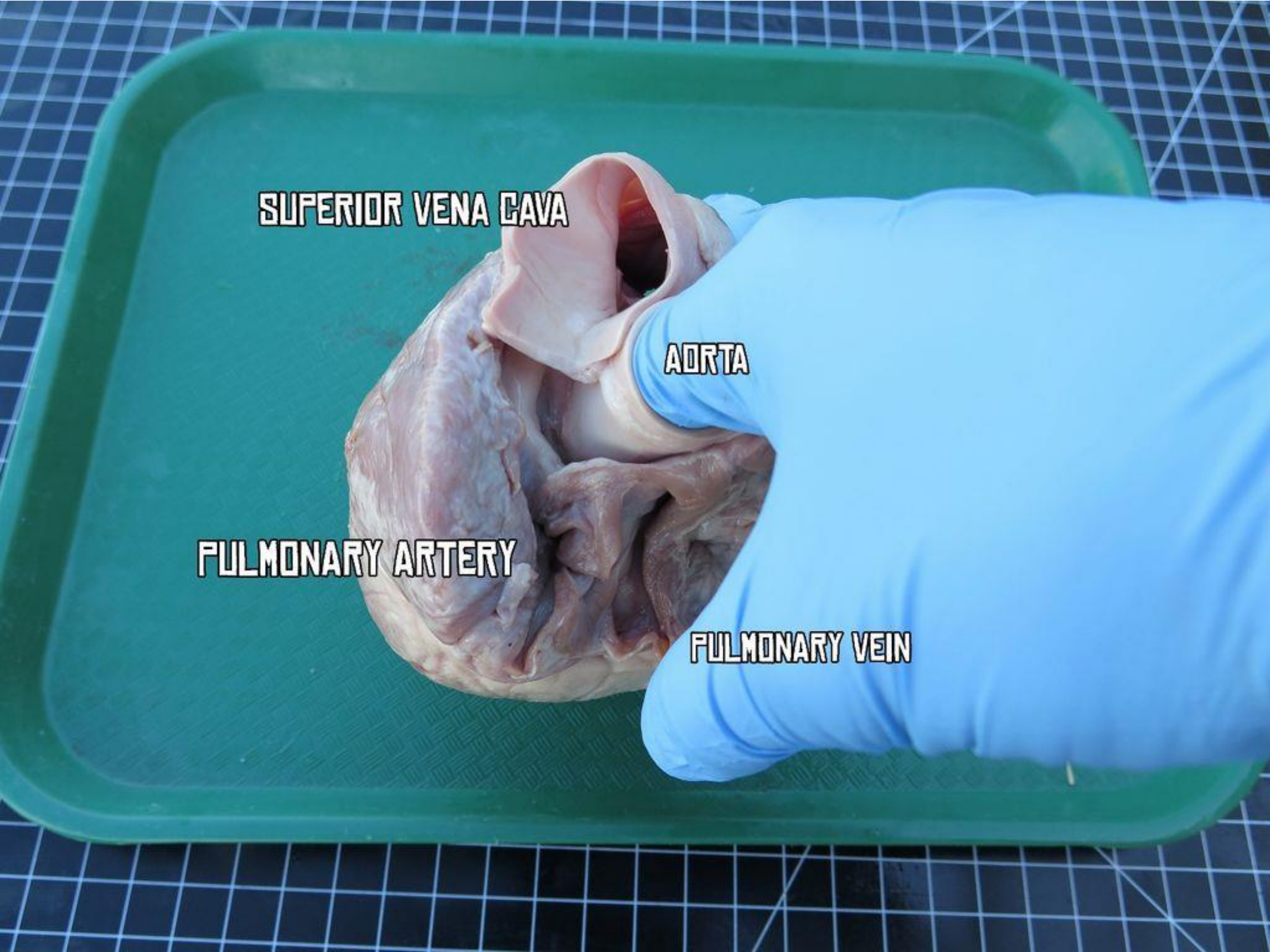
Name four parts of the heart.

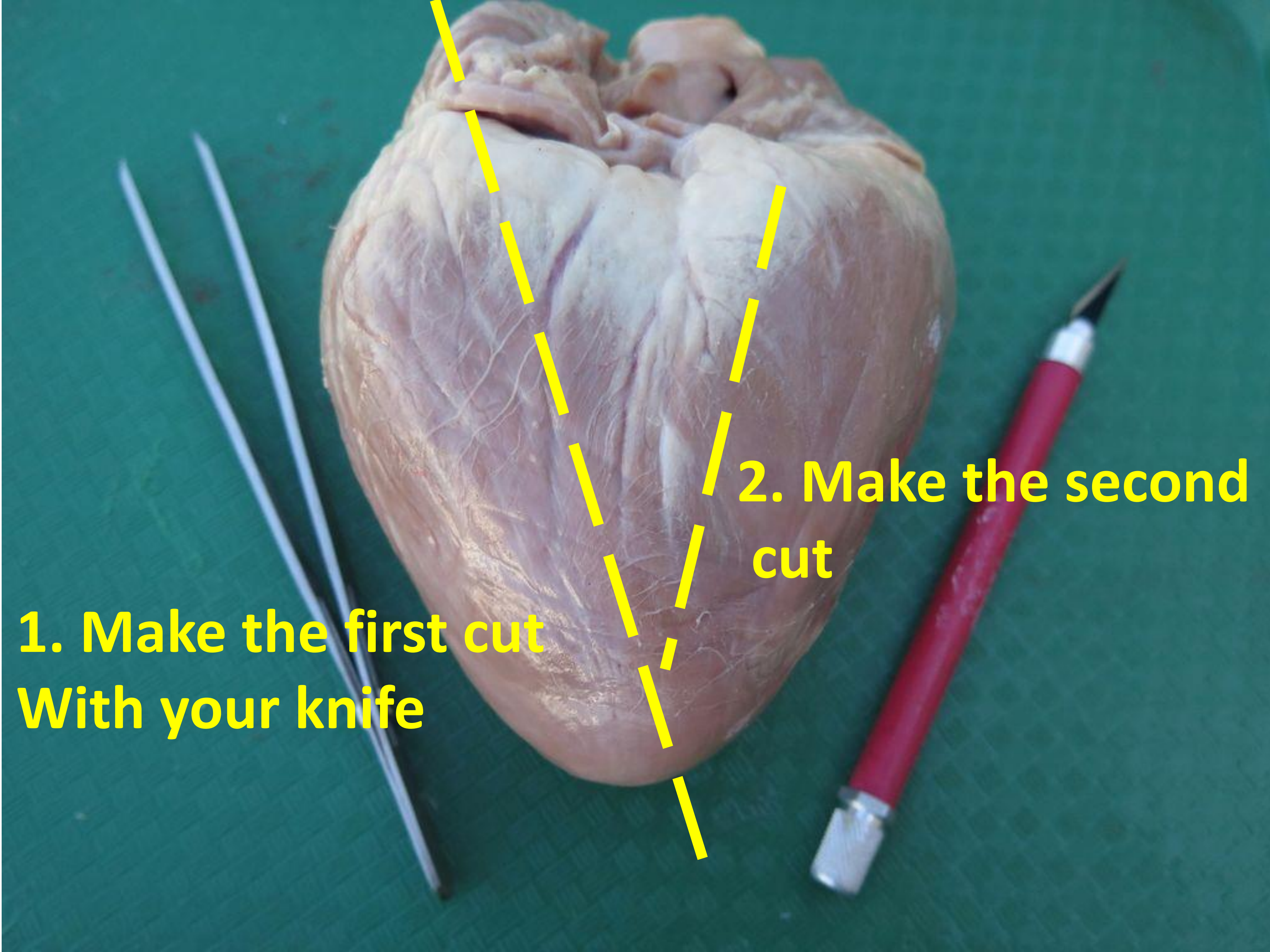
State where the aorta sends blood

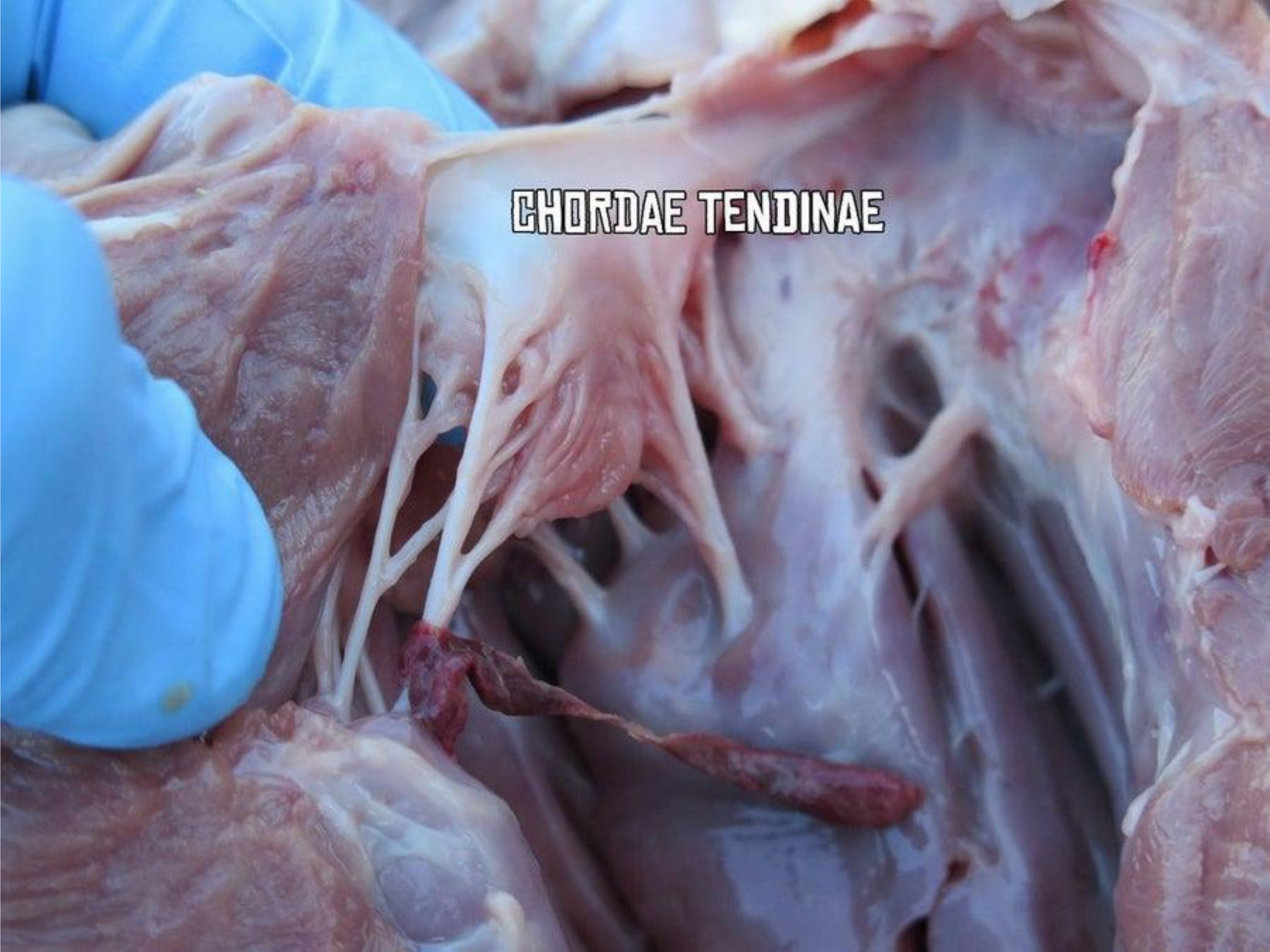
Lay the heart on the table like this.



Try to find the four vessels of the heart and stick a lollipop stick down each one.


SUPERIOR VENA CAVA
AORTA
PULMONARY ARTERY
PULMONARY VEIN


2. Make the second
cut
1. Make the first cut
With your knife

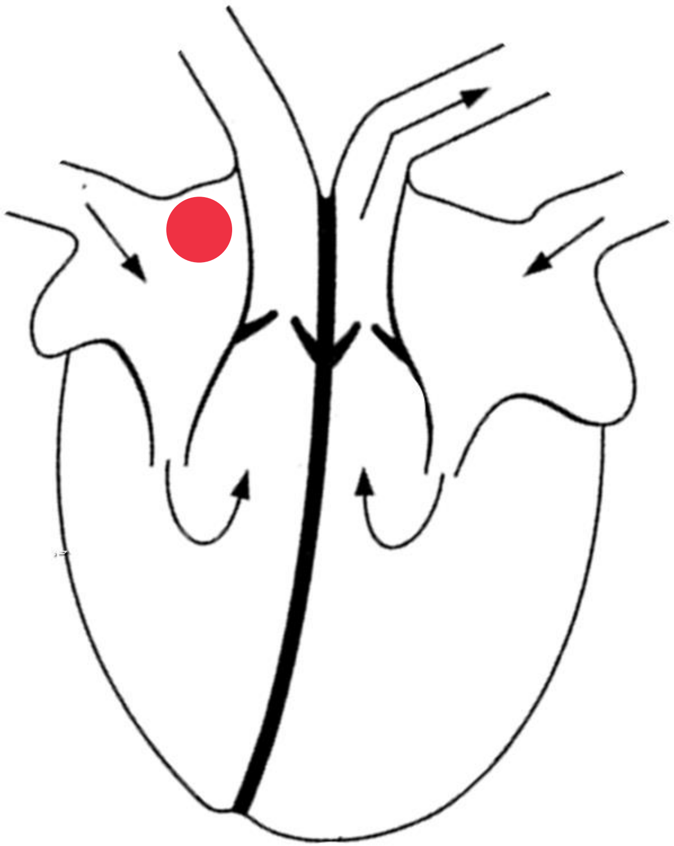

CHORDAE TENDINAE

The heart has a natural resting rate which we measure in beats per minute (bpm).

Resting heart rate is controlled by the **pace maker** which are a group of cells in the **RIGHT ATRIUM.**

Average resting bpm

Elite athletes 30-50 bpm
Healthy 60-80 bpm
Unhealthy 90+ bpm



TASK: Find your pulse and lets test your resting heart rate

If the pacemaker stops working it causes an abnormal rhythm this is called an arrhythmia

Abnormal rhythm (arrhythmia)

Too Fast

- PROBLEM -

Can't pump blood properly

Treatment

Ablation → removing faulty pathways

Defibrillator → Electrical Shock to restart heart

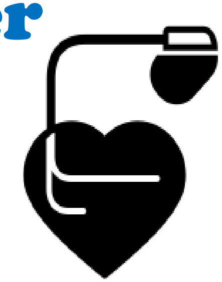
Too slow

- PROBLEM -

Not enough oxygen pumped to the body

Treatment

Pacemaker



Activity: Watch this video about pacemakers. Write down what you learn

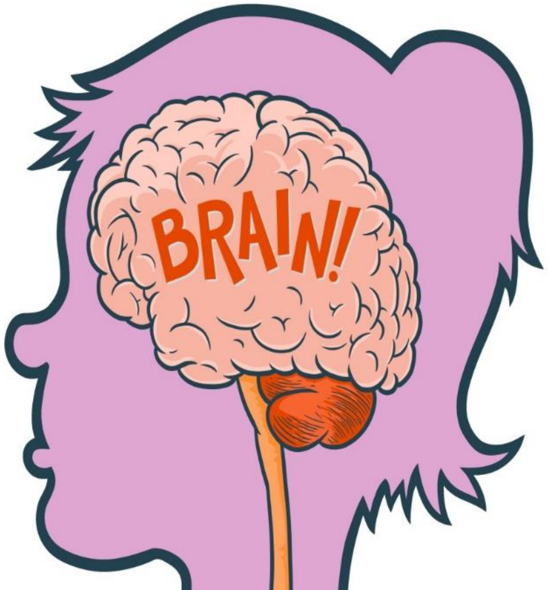
WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!

Review: What did you learn?



Review: What did you learn

Describe it? Small, light weight, electrical device with wires extending into the heart

Where does it go? Main part under skin. Wires extend into the heart

Explain what it does? Sends strong regular electrical signals to your Heart to stimulate it to beat properly.

Is it a perfect system? Can have lead failure and can require open heart surgery to get it replaced in 2% of cases.

Advantages of new pacemakers? No need for wires so can last For a much longer time.